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ORIGINAL ARTICLE

Bacteriophages: A possible solution to combat enteropathogenic *Escherichia coli* infections in neonatal goats

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Significance and Impact of the Study: Ruminants are susceptible to various bacterial infections, which are treated with antibiotics that are increasingly failing due to AMR. Hence bacteriophages can be employed to treat such bacterial infections. Due to rising AMR the interest in phage therapy has recently been renewed, although its application in veterinary medicine remains a long way off. This research focused on the management of enteropathogenic *E. coli* (EPEC) through optimal PCR-based detection strategy and the use of bacteriophages as a therapeutic alternative to antibiotics to combat EPEC infections in neonatal goats.

Keywords
enteropathogenic *Escherichia coli* (EPEC), *Stx*1 gene, bacteriophages, antimicrobial resistance (AMR), multi-drug resistant (MDR).

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Abstract

Due to awareness and benefits of goat rearing in developing economies, goats' significance is increasing. Unfortunately, these ruminants are threatened via multiple bacterial pathogens such as enteropathogenic *Escherichia coli* (EPEC). In goat kids and lambs, EPEC causes gastrointestinal disease leading to substantial economic losses for farmers and may also pose a threat to public health via the spread of zoonotic diseases. Management of infection is primarily based on antibiotics, but the need for new therapeutic measures as an alternative to antibiotics is becoming vital because of the advent of antimicrobial resistance (AMR). The prevalence of EPEC was established using bfpA gene, uspA gene and Stx1 gene, followed by phylogenetic analysis using Stx1 gene. The lytic activity of the isolated putative coliphages was tested on multi-drug resistant strains of EPEC. It was observed that a PCR based approach is more effective and rapid as compared to phenotypic tests of Escherichia coli virulence. It was also established that the isolated bacteriophages exhibited potent antibacterial efficacy in vitro, with some of the isolates (16%) detected as T4 and T4-like phages based on gp23 gene. Hence, bacteriophages as therapeutic agents may be explored as an alternative to antibiotics in managing public, livestock and environmental health in this era of AMR.

Introduction

Escherichia coli is known to be a natural commensal of warm-blooded animal gut flora. A new and more virulent generation of the multi-drug resistant (MDR) bacterial strain has been developed due to the indiscriminate use of antibacterial agents (Magiorakos *et al.* 2012). The pathogenicity of *E. coli* depends on its virulence factors, invasins, heat-labile and heat-stable enterotoxins, verotoxins and adhesin or colonization factors (Relman and Falkow 2015). It is pathogenically classified into diarrhoeagenic *E. coli* and uropathogenic *E. coli*.

diarrhoeagenic *E. coli* is further subdivided into enterotoxigenic *E. coli* (ETEC), enteropathogenic *E. coli* (EPEC), enteroinvasive *E. coli* (EIEC), enteroaggregative *E. coli* (EAggEC) and enterohemorrhagic *E. coli* (EHEC), which causes diarrhoea and other associated illnesses (Garrine *et al.* 2020). *Escherichia coli* are present in both humans and livestock, where ruminants (sheep and goats) have been identified as a substantial pool and human infection source (Rahman *et al.* 2020). Small ruminants, such as goats and lambs, are more prone to gastrointestinal illnesses (Bayne and Edmondson 2021). Pathogenic variants of *E. coli* affect humans and animals, causing diseases worldwide (Croxen *et al.* 2013), where EPEC is primarily responsible for fatal diarrhoea in developing countries in infants (Bugarel *et al.* 2011; Mare *et al.* 2021). Cattles are the natural reservoir of O157 *E. coli* containing verotoxin (Lim *et al.* 2010), and the ingestion of unpasteurized milk presents a high risk of O157: H7 *E. coli* infection (Disassa *et al.* 2017). Ruminant animals are conceptualized to pose as primary reservoirs of shiga toxin (*Stx*) producing *E. coli* (STEC), where STEC infection resulted in 1 million diseases and 100 fatalities in 2010, as reported by the World Health Organization (Havelaar *et al.* 2015; Kim *et al.* 2020). An outbreak was reported in the United States where 25 people got infected due to STEC O157: H7 resulting in 1 death linked to leafy greens (CDC 2018).

India ranks second in goat population with nearly 148·9 million approximately (Lata and Mondal 2021). In India, goats are considered poor people's banks or insurance schemes, distributed between landless and poor farmers, but the rapid surge in the animal farming sector increased commercial goat farming (Kumar *et al.* 2010). Goat rearing in India has enormous demand as they are the main meat (chevon) producing animals (Mazhangara *et al.* 2019). Goat farming has been gaining traction in the commercial and domestic sectors over the last few years due to its popularity. However, as goat meat and milk are generally recognized worldwide, constraints on the growth of goat enterprises have also been observed due to its association with bacterial and parasitic infections (Ayaz *et al.* 2018; Miller and Lu 2019) that are critical for causing human infections and may pose a threat to public health by spreading zoonotic diseases (Monteiro *et al.* 2018). A retrospective study on goats from 1988 to 2012 showed that 43·67% of deaths were due to enteritis, followed by gastrointestinal parasitism and gastric diseases (Pawaiya *et al.* 2017). Currently, medical science is entirely dependent on antibiotics to treat infectious diseases and strives to keep the human race healthy and protect animals. The fact that infectious diseases are no longer treatable with antibiotics due to antimicrobial resistance (AMR) portends a bleak future for the healthcare system, which is extremely concerning for our society.

An increase in drug-resistant bacteria has contributed to significant morbidity and mortality due to the inappropriate use of antibiotics (Fair and Tor 2014). As a result of these circumstances, scientists and the pharmaceutical sector are rekindling their interest in the use of bacteriophages. The utilization of broad-spectrum drugs affects the gut microbial community and its composition (Langdon *et al.* 2016). Structural modifications contribute to changes in genes' expression, the protein's activity and the gut's overall metabolism, the results of which may or may not be temporary (Hasan and Yang 2019). Bacteria whose genes are altered by broad-spectrum antibiotics may serve as a gene pool for such resistant bacteria, contributing to the spread of antibiotic resistance (Peterson and Kaur 2018). This has prompted researchers to rethink a decades-old phage therapy approach as a viable new therapeutic alternative against bacterial infections. A study was conducted where antimicrobial activity was exhibited by phage mixture against *S. aureus* from bovine mastitis (Titze and Kromker 2020). Since bacteriophages are highly selective, they are ineffective against the host nonpathogenic flora (Principi *et al.* 2019). Thus, phages may be one of those multi-strategic instruments that could modulate microbial diversity and assist us in combating MDR bacterial strains as we are about to reach the post-antibiotic period, where we are lagging in the fight against many diseases due to our same old classical approach (Aslam *et al.* 2018; Bhargava *et al.* 2021; Raza *et al.* 2021). In this article, we describe the isolation of highly pathogenic strains of *E. coli* and their molecular characterization in relation to their AMR traits that potentially constitute a risk. We also isolated appropriate coliphages and tested their bactericidal efficacy on indicator species to identify prospective phage therapy candidates for the near future.

Result and discussion

Antibiotics susceptibility testing

The isolated strains were MDR but non-ESBL producing, where the majority were resistant to amoxy-clavulanic acid, norfloxacin, cefepime, meropenem and nitrofurantoin and were sensitive to ampicillin and gentamycin.

Molecular confirmation and characterization of *E. coli* isolates

Around 28 kids (87%) showed the presence of EPEC by *bfp*A gene-based SYBR green real-time PCR (Fig. 1), *bfp*A gene-based conventional PCR (Fig. 2a) and *usp*A gene

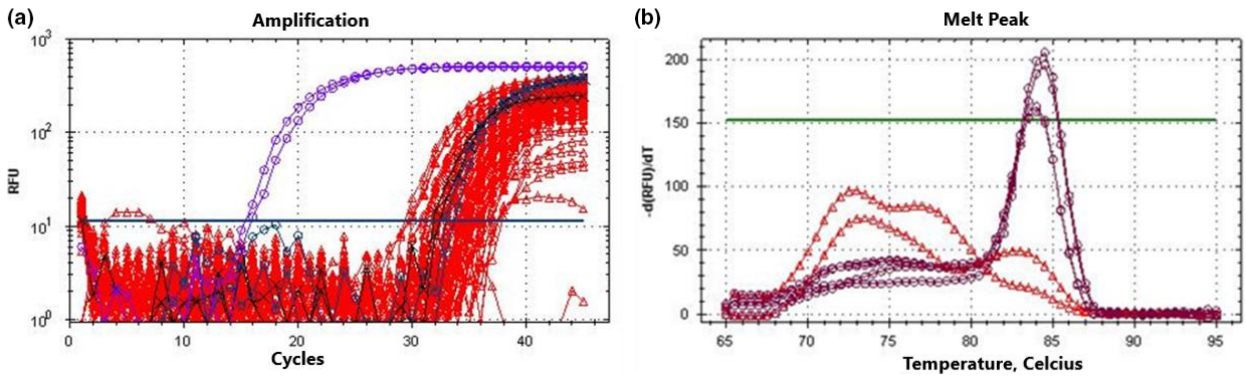

Figure 1 *bfp*A gene SYBR green real-time PCR for EPEC screening: (a) Cycle quantification plot (Log scale) for *bfp*A gene amplification (Purple—positive control, red—unknown), (b) melt peak for the specific amplified product of 158bp.

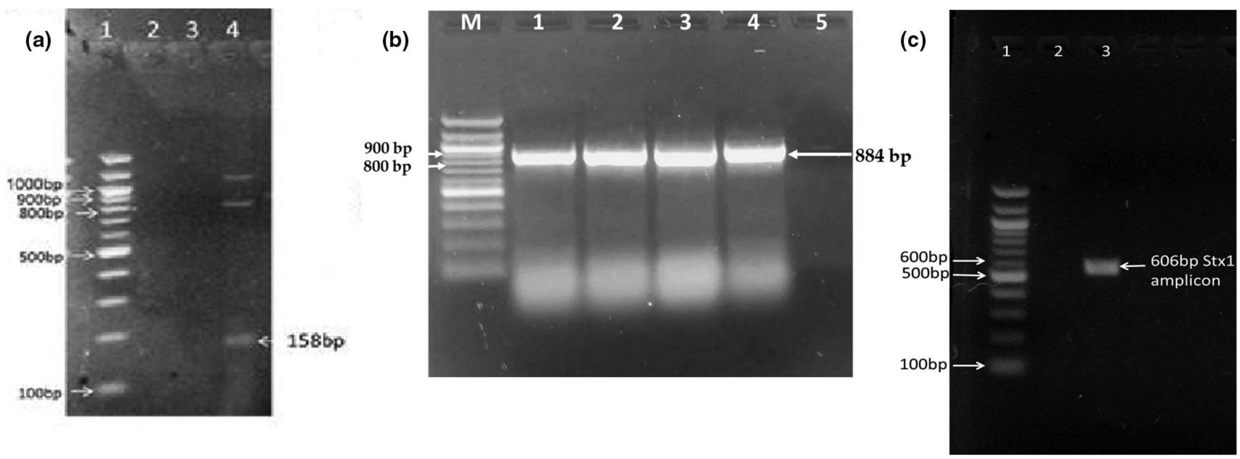

Figure 2 (a) *bfp*A gene conventional PCR for EPEC. Wells, 1: 100bp DNA ladder, 2: No template control, 3: Negative control, 3: 158bp *bfp*A gene amplicon, (b) *usp*A gene PCR for *E. coli*. Wells, M: 100bp DNA ladder, 1, 3–7: 884bp *usp*A gene amplicon, 2: No template control, (c) Gel picture showing the *stx*1 gene amplification of the CIRG1: 2017 Shiga toxin-producing *Escherichia coli* (STEC) isolate (Lane 3) against no-template control (Lane 2) and 100bp DNA ladder (Lane 1).

conventional PCR (Fig. 2b). Out of all, one isolate of *E. coli* was *stx*1 positive by conventional PCR (Fig. 2c). The CIRG STEC strain showed 99% homologous with other reference strains.

Phylogenetic analysis was conducted using the minimum evolution (ME) method (Fig. 3a) of MEGA6 software (Tamura *et al.* 2013). It computed the optimal tree with the branch length sum equal to 0·05187866. The percentage of clustered replicate trees and the related taxa are subject to the bootstrap test (500 replicates). Evolutionary distances have been evaluated using the Maximum Composite Likelihood method, and the number of base substitutions per site is in units. The ME tree was scanned at a search level of 1 utilizing a close-neighbour-interchange (CNI) algorithm. To construct the initial tree, the Neighbour-joining algorithm was used. The evaluation included 13 sequences of nucleotides. The positions included in Codon were 1st+2nd+3rd+Noncoding. It excluded all positions containing gaps and missing data. In the final dataset, there were 520 positions overall.

Based on phylogenetic analysis, the STEC CIRG1:2017 strain was present in one of the two major branches with close association with STEC strains, including ONT: H34, 4756/98, O6E01767 reference strains belonging to various subclasses. The strain 4756/98 was present in the same clade where STEC/CIRG: 2017 was placed. However, the DEC 10J strain was present in the other branch and SWUN 4124 and FD930 strains.

A sequence identity plot constructed using BioEdit V. 7.2.5 (Hall 1999) of various *E. coli* mapped for nucleotide variations is presented below (Fig. 3b). The most common point mutations observed in some strains, including CIRG1: 2017 *Stx*1, are G → A at the 1511 nucleotide position compared to *E. coli* strain' N1508 *Stx*1.'

Figure 3 (a) Phylogenetic analysis of *Stx*1 gene of STEC strain CIRG1: 2017 with other standard reference cultures. The evolutionary history of *the stx*1 gene using the Minimum Evolution method, (b) Illustration showing sequence identity plot for *stx*1 gene of various STEC reference strains with the CIRG1:2017 strain. The dotted line represents homology with the first sequence. The nucleotides against the dotted line represent mutations in the sequence.

Isolation of bacteriophage cocktail and its strains

Several cocktails were isolated from ghats along the Ganges in Varanasi, and the bacteriophage strains were separated using the overlay method (Kropinski *et al.* 2009). Collected water specimen, when initially dropped on 3-h-old lawn culture of EPEC strain, ambiguous drops are apparent after 24 h of incubation, which becomes more comprehensible as the day progresses when the technique is repeated numerous times for successive five days. When the cocktail was implemented to overlay method, distinct morphological plaque of different sizes (small, medium and large) was observed (see Fig. S3).

In vitro activity of phage (n = 50)

Phage isolated against one specific EPEC strain was later monitored for its activity against a variety of other isolated EPEC bacterial strains. Phages ($n = 50$) that were dropped on 3-h-old lawn culture cleared the area where they were dropped. When the phage cocktail was tested against the other test bacteria (EPEC), it demonstrated a decrease in bacterial growth after 24 h and cleared the entire plate when the phage lysate was employed at higher concentrations. Thus, our research demonstrated the efficacy of prepared phage cocktails against several bacterial strains in vitro, as well as promising results against MDR bacteria.

Phage stability study

When the lytic activity of phages (Φ EC3, Φ EC4, Φ EC13, Φ EC15 and Φ EC20) was assessed at different pH conditions by exposing them to varying ranges of pH (3, 6, 7, 8, 10 and 12) for 48 h, it was found that phages were viable at pH 3, 6, 7, 8 and 10 got inactivated at pH 12. When it came to thermal tolerance, all five phages (Φ EC3, Φ EC4, Φ EC13, Φ EC15 and Φ EC20) showed lytic activity up to 45 and at higher temperature reduced or no lytic activity was observed.

One-step growth curve analysis

The latent period and burst size of phages (EC3, EC4, EC13, EC15 and EC20) were determined in TMG at 28 by one step growth curve test. The latent period was approximately 45 min with the burst size of about 148 PFU per infected cell (55 min) for Φ EC3 (Fig. 4a), 50 min latent period with a burst size of 78 PFU per infected cell (60 min) for Φ EC4 (see Fig. S4), 35 min latent period with the burst size of 120 PFU per infected cell (55 min) for Φ EC13 (see Fig. S5), 40 min latent period with a burst size of 104 PFU per infected cell (50 min) for Φ EC15 (see Fig. S6) and 40 min latent period with a burst size of 65 PFU per infected cell (55 min) for Φ EC20 (see Fig. S7). Owing to the large burst size and relatively short latent period of Φ EC3 makes it favourable for its potential use in bacteriophage therapy.

Bacterial growth inhibition at different MOIs

After 80 min of incubation, the highest MOI 1·0 resulted in the maximum lytic activity, with EPEC titre decreasing from 10^8 CFU ml^{-1} to around 10^7 CFU ml^{-1}. Furthermore, a reduction in bacterial titre was seen for MOIs 0·1, 0·01 and 0·001 as compared to the initial bacterial titre at 0 min (Fig. 4b). However, for MOIs 0·1, 0·01 and 0·001, the bacterial titre remains constant until 120 min, whereas for MOI 1·0, the bacterial titre increases after 100 min. As a result, using phage at greater concentrations resulted in a rapid reduction of bacterial titres, whereas lower MOIs resulted in a moderate reduction of bacterial titres (Yazdi *et al.* 2020).

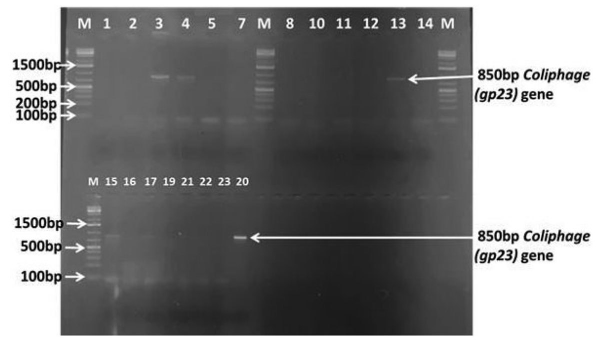

Figure 5 Lanes, M-Marker, 1 to 20 are the lanes corresponding to the bacteriophage isolates ΦEPEC1 to ΦEPEC20 respectively. Lanes 3, 4, 13, 15 and 20 are bacteriophage isolates that were positive for *gp*23 amplicon suggestive of T4 and T4-like phages by PCR.

Molecular characterization and confirmation of T4 and T4-like phages screening by major capsid protein *gp23 gene PCR*

From the total bacteriophage isolates (n = 50) screened, eight isolates (16%) amplified the 850 bp amplicon suggestive of the major capsid protein *gp*23 gene of T4 and T4-like phages by PCR. The 1·5% TAE-Agarose gel stained by EtBr is shown below (Fig. 5).

Diarrhoea in neonatal goats caused by EPEC is one of the major factors of high morbidity and mortality in developing nations (Singh *et al.* 2018). The present study explores the prevalence of enteric diseases and their detection via different molecular techniques, which, under field conditions, are a significant cause of mortality in neonatal infants (Pawaiya *et al.* 2017). Neonatal enteritis is caused by various etiological agents, with the incidence of mixed infections complementing each other and amplifying the condition due to immunity imbalance (Singh *et al.* 2018). EPEC is a bacterium with many serotypes, of which only

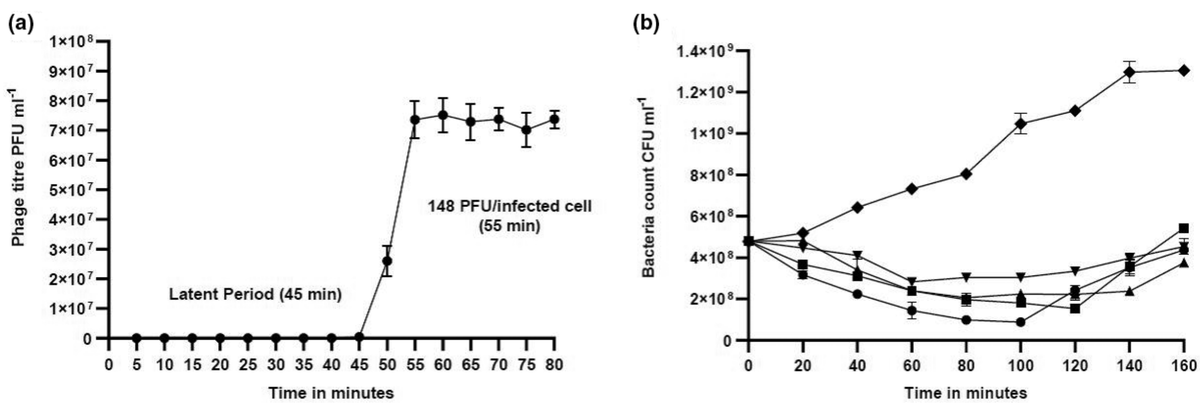

Figure 4 (a) Phage EC3 one-step growth curve in the presence of enteropathogenic *Escherichia coli* (EPEC) as a host, (b) Inactivation of EPEC by phage EC3 at MOIs 1·0 (●), 0·1 (■), 0·01 (▲) and 0·001 (▼) during 160 min and control represent bacteria control (♦). Values for both the graphs represent the mean of three independent assay and error bars represent standard deviation.

some are pathogenic, causing diarrhoea and septicaemia, which results in the death of goat kids if left untreated. EPEC isolate detection serves as an indicator for the presence of virulent *E. coli* in the herd, detected via *bfp*A gene-based SYBR green real-time PCR. The more adequate, sensitive and rapid approach for detecting diarrheagenic *E. coli* strains of veterinary importance is the PCR-based approach compared to phenotypic tests conducted on individual colonies. The most widely used chemistries available for real-time PCR assays are SYBR Green-based assays (Tajadini *et al.* 2014). In general, compared to traditional diagnostic methods, including isolation and culture techniques, molecular tests are rapid, sensitive, and less tedious (Franco-Duarte *et al.* 2019). Early identification, diagnosis, and development of a vaccine would help protect neonatal goats, effectively preventing farmers' economic losses. EPEC infection in neonatal goats leads to severe, acute diarrhoea coupled with severe dehydration, extreme acid-base and electrolyte imbalance, and mild diarrhoea mortality without systemic disease, often in less than 12 h (Gruenberg 2014).

Various forms of antibiotics are now indiscriminately used worldwide in the veterinary industry to encourage livestock growth and care (Sachi *et al.* 2019). Globally, 63 151 ± 1 560 tonnes of antibiotics are used in livestock every year, which are expected to rise by 2030 (Tiseo *et al.* 2020). Antibiotics are used in animal husbandry for therapeutic and prophylactic purposes, the global consumption of which is double in animals relative to humans (Kümmerer 2008; Aarestrup 2012). Many studies have shown that significant portions of antibiotics are released into the environment without modification, that is, with possible antimicrobial activity (Kümmerer 2008). The requirement for animal protein (milk and meat) is increasing worldwide, facilitating the expansion of industrial farming and, with the injudicious use of antibiotics for treatment purposes, contributes to AMR (Sachi *et al.* 2019; Tiseo *et al.* 2020). One of the most common priority areas recognized by national and international agencies is AMR which is proliferating as a silent pandemic (Sharma *et al.* 2018). Hence, a renewed focus on research investments is needed to identify alternative, secure, cost-effective and creative solutions in parallel with discovering new antibiotics. One such biological entity that can be employed against antibiotic-resistant bacteria is bacteriophages (Saussereau *et al.* 2014). Bacteriophages can infect and kill prokaryotes without any adverse effect on eukaryotes and are highly specific (Principi *et al.* 2019). Further, in the current study, we subjected the coliphages for screening of T4 and T4-like phages of using *gp23* (Major capsid protein) gene to understand the type of bacteriophage isolated from the sources. In the past, many authors reported the effective use of phages against pathogenic *E. coli* causing diarrhoea in neonatal calves, pigs, lambs and poultry (Smith and Huggins 1983; Johnson *et al.* 2008; Anand *et al.* 2015). In the current study, besides the abundant EPEC strains, one strain was identified and molecularly characterized as *Stx*1 producing. Besides isolation of phages, assessment of its efficiency and virulence, the characterization is also equally critical in the development of a phage therapeutic preparation (Johnson *et al.* 2008). While carrying the double-edged sword of AMR and virulence genes, an alternative approach is required, and hence the bacteriophages screened by PCR were tested for efficiency against such pathogenic strains of *E. coli*. To bolster the preceding statement, we isolated phages and tested them against MDR EPEC strains, the in vitro analysis of which revealed complete clearance of the bacterial lawn in 120 h. Also, if they are employed for *in-vivo* treatment in goats, we isolated five different strains of phages that can be made into a cocktail and administered orally to the goats in which drugs are ineffective. Future research can be recommended based on the finding that phages can be used *in-vivo* and options for concentrations and different administration routes.

Highly virulent AMR *E. coli* strains are a 'double whammy' that necessitates a different strategy. Hence, in the current study, we did a thorough characterization of the isolates targeting pathogenic genes and screened more isolates for enteric pathogenicity by *bfp*A gene-based SYBR green real-time PCR. One isolate, which carried both enteropathogenic (*bfp*A) as well as enterotoxigenic (*stx*1), was used as a model isolate (CIRG1:2017) for assessing the potential of isolated coliphages. Furthermore, the same isolate was sequenced for the *stx*1 gene and found the G → A mutation and the phylogenetic tree showing taxonomically close to German origin strain *E. coli* strain 4756/98 (Zhang *et al.* 2002). In this scenario, it is further required to screen more isolates that carry the virulence factors without compromising AMR characteristics detection. A more detailed and extensive study would help devise a strategy in deciphering the AMR and virulence genes transmission dynamics in *E. coli* or other vital pathogens in the future. Nevertheless, the judicious use of antibiotics coupled with venturing into alternate therapeutics could help combat the menace of AMR strains that are fuelled with virulence factors putting livestock production at risk and aggravating the zoonotic flow.

Material and methods

Sample collection and isolation of *E. coli strains from the goat*

A study was conducted in hebdomadic goat kids that showed acute diarrhoea with pasty greenish faecal matter soiling the perianal region. Faecal swabs were collected from 32 goat kids born between September–November (kidding season) from various unorganized herds in an organized goat farm. For bacteriological culture, swabs were washed with 1·0 ml of sterile PBS and vortexed before inoculation to culture media. They were cultured in MacConkey's agar (MCA) and Eosin Methylene blue (EMB) agar and incubated at 37 . The colonies obtained of *E. coli* were gram stained, and specific biochemical tests were conducted. The lactose fermenting colonies of MCA were further selectively streaked on Congo-Red dye agar and incubated at 37 for 72 h, which resulted in the development of colonies with a brick red colour.

Molecular characterization of *E. coli by PCR*

DNA extraction was performed using the Nucleopore® DNA Kit (Genetix), implementing the manufacturer's protocol from pure sub-cultured MCA colonies. The DNA is then used to detect EPEC and STEC based on the conventional PCR (cPCR) detection using *bfp*A and *stx*1 genes respectively. The cPCR primers for *bfpA* were the same as the SYBR green real-time primers designed in-house in the laboratory (given below in next para), while the *stx*1 gene was amplified using *stx*1F: 5'CACAATCAGGCGTCGCCAGCGCACTTGCT3' and *stx*1R: 5'TGTTGCAGGGATCAGTCGTACGGGGATGC3' (Talukdar *et al.* 2013). The identification of *E. coli* molecularly was made by PCR amplification of the universal stress protein A (*usp*A) gene utilizing species-specific primers (F-5'-CCGATACGC TGCCAATCAGT-3' & R-5- ACGCAGACCGTAGGC CAGAT-3') (see Fig. S1). The annealing temperature was kept at 55 for 1 min. The amplified *stx*1 gene was sequenced by Sanger's dideoxy method using the BigDye terminator kit, and the raw chromatogram data generated was processed using BioEdit 7.2 tool for proof-read and alignment (Hall 1999). Further sequence identity plot was done to compare the nucleotide composition and point mutations in the coding region of the *stx*1 gene of the above-isolated strain. The clustal-W aligned sequences were exported to MEGA 6.0 for phylogenetic analysis by comparing with the Stx1 sequences of other reference strains reported elsewhere (Tamura *et al.* 2013). Similarly, the DNA samples were also used for screening of EPEC by using *bfp*A SYBR green real-time PCR. The EPEC colonies with brick red colour in Congo-Red dye agar isolated were also used for molecular screening by *bfp*A SYBR green real-time PCR.

Molecular screening of EPEC using *bfp*A gene-based SYBR-green real-time PCR.

Bundle forming pilin protein A gene primers: Size of product 158bp.

*bfp*A F: 5′-ATGGTGCTTGCGCTTGCTGC-3′,

*bfp*A R: 5′-AATCCACTATAACTGGTCTGC-3′

Antibiotic susceptibility testing of isolated bacterial strains

Isolated EPEC strains were put up for antibiotic susceptibility based on the modified Kirby Bauer Disk diffusion method, as per the CLSI (Clinical and Laboratory Standard Institute) guidelines (Bauer *et al.* 1966). Set of antibiotics used, include amikacin (30 μg), amoxy-clavulanic acid (20/10 μg), ampicillin (10 μg), ceftriaxone (30 μg), ceftazidime (30 μg), ceftazidime-clavulanic acid (30/10 μg), ciprofloxacin (5 μg), co-trimoxazole (1·25/23·75 μg), cefotaxime (30 μg), gentamicin (10 μg), norfloxacin (10 μg), aztreonam (30 μg), cefepime (30 μg), meropenem (10 μg), piperacillin-tazobactam (100/10 μg), nitrofurantoin (300 μg), colistin (10 μg). In vitro presence of extended-spectrum beta-lactamase (ESBL) was also confirmed according to CLSI guidelines by using cephalosporin (ceftazidime) and their respective cephalosporin/clavulanate (ceftazidime-clavulanic acid) discs (Paterson and Bonomo 2005).

Collection and isolation of bacteriophages

For the isolation of bacteriophages, water samples were collected from multiple ghats of the river Ganges of Varanasi from multiple locations in a sterile plastic container. First, a sample from the water specimen was treated with 1% chloroform (v/v) for 10 min with continuous vortexing/inversion for bacteriophage cocktail isolation. It was later centrifuged for 10 min at 10 778 ***g***. Next, the supernatant collected was flooded on the 90 mm Mueller-Hinton Agar (MHA) lawn culture (exponentially grown 4 h old EPEC culture with an optical density (OD) at 600 nm = 0·6). It was incubated overnight at 37 . After incubation, the lawn culture was washed with TMG (Tris HCL, Magnesium Sulphate, Gelatin pH 7·4) buffer and treated with 1% chloroform followed by vortexing/inverting for 10 min. Soon after the chloroform treatment, it is centrifuged for 10 min at 10 778 ***g***. Finally, the supernatant is transferred carefully to another properly autoclaved 1·5 ml centrifuge tube without disturbing the sedimented lysed bacteria. The whole process of centrifugation was repeated four times. Again, a 4 h old lawn culture of the bacterial host (EPEC) was prepared; the supernatant collected was dropped on the plate and was incubated for 24 h at 37 . All of the procedure was repeated multiple times till the complete clearance of the lawn culture was obtained. The supernatant collected was preserved at 4 for further use.

Isolation of different bacteriophage strains from cocktails

The isolated bacteriophage cocktail was tested for broad-spectrum lytic activity against all the other isolated EPEC strains and filtered through a 0·22 μm pore size syringe filter to remove contaminants from the bacteriophage mixture. The soft agar overlay method was used to segregate phage strain from bacteriophage cocktail (Kropinski *et al.* 2009). The soft agar (0·8% agar) is kept at a molten state at a temperature of not more than 40–43°C in the water bath. Bacterial (20 μl of 1 McFarland standard) and phage (200 μl) suspension mixed with TMG buffer (780 μl) is added to the soft agar, which is further overlaid on the MHA plates and incubated for 18–24 h at 37 . After incubation, different morphological plaques of different sizes were observed against the host bacterium, which was later cut out and put into Luria-Bertani (LB) broth and was incubated overnight at 37 for further purification. The different numbers of plaques cut out after incubation were treated with 1% chloroform (v/v) for 10 min and were centrifuged at 10 778 ***g*** for 10 min. Then, the supernatant obtained was transferred to another 1·5 ml centrifuge tube, and centrifugation was repeated three more times. The spot test method was conducted where the exponentially grown bacterial lawn culture was prepared on MHA plates, and 5–10 μl of phage filtrate was dropped as a spot for evaluation of its lytic activity. Initially, the bacterium used to isolate the phage was assumed to be a phage's host bacterium whose lytic activity was further evaluated against other isolated EPEC strains.

Purification of phage lysates

For phage replication, high titres of isolated phage lysates were prepared (propagation on host bacterium) and were put up for dialysis. Several passages were carried out over three days to assess phage lytic activity, which resulted in a rise in phage concentration. The obtained high titre phage lysates were pocketed into the dialysis membrane pouch, were precipitated with 20% PEG 8 000 (polyethylene glycol) and 2·5 NaCl. The obtained phage particles were washed five times with PBS (phosphate buffer saline; pH-7·4; for 100 ml: 0·8 g, NaCl; 0·02 g, KCl; 0·142 g, Na_2HPo_4; 0·24 g, KH_2Po_4) for three consecutive days by changing buffer every 8 h and were stored at 4 for 24 h. For further analysis, the distilled phage suspension was stored at 4 .

Phage stability analysis

The stability of phages (cocktail) was analysed at different pH and temperatures. For thermal stability tests, phage lysates were incubated for 48 h at 4, 15, 21, 37, 45, 55 and 65 in a temperature-controlled water bath. For pH, stability tests were performed using TMG buffer, and pH was adjusted using 1N NaOH and 1N HCl. In addition, the phage lysates were incubated at different pH 3, 6, 7, 8, 10 and 12 for 48 h, which was further tested for their lytic activity using the spot test method.

One-step growth curve

The burst size of phages EC 3, EC 4, EC 13, EC 15, and EC 20 was determined using a one-step growth curve experiment (Adams 1959). In 15 ml LB broth, an EPEC strain (0·9 ml) was cultured with a final concentration of 10^{11} CFU ml^{-1}. The phages (0·1 ml) were added to a fresh bacterial culture in LB broth at a final concentration of 10^9 PFU ml^{-1} to achieve an MOI (multiplicity of infection) of 0·01, and the entire suspension was incubated for 10 min at 28°C without shaking so as to allow phage adsorption into the bacterial cell. The bacteria and phage suspension were centrifuged at 10 000 ***g*** for 5 min, the supernatant was discarded, and the pellet was washed in 15 ml TMG and incubated at 28°C. Throughout the incubation period of up to 80 min, samples (1 ml) were harvested at 5 min intervals and immediately tittered MOI using the soft agar overlay method (Kropinski *et al.* 2009). The mean numbers of PFU ml^{-1} were exhibited along with standard errors of the means in three independent assays.

Killing curve of bacteria at different MOIs

Bacterial inhibition assays were carried out with various MOIs for the phage with a big burst size and a short latent period with minor modifications (Yazdi *et al.* 2020; Menon *et al.* 2021). First, EPEC culture was suspended in 10 ml LB broth with a final concentration of 108 CFU ml^{-1}. Next bacterial suspension (90 μl) was combined with phage (10 μl) at varied MOIs of 1·0, 0·1, 0·01 and 0·001 (with concentrations of 10^8, 10^7, 10^6 and 10^5 PFU ml^{-1} respectively), followed by an 18-h incubation at 28°C. A bacterial control without phage was utilized for each independent assay, and aliquots were taken after 0, 20, 40, 60, 80, 100, 120, 140 and 160 min of incubation, with bacterial concentration quantified by OD at 600 nm. All of the independent assays were performed in triplicates.

Screening of coliphages of the *Myoviridae family using the gp23 gene*

The bacteriophage strains (n = 50) isolated from various sources were screened for the various T4 and T4-like phages of the *Myoviridae* family targeting the major capsid gene *gp*23. Following the manufacturer's instructions, each phage sample was subjected to DNA extraction using the Wizard® genomic DNA kit (Cat# A1120, Promega).

The resultant DNA was quantified using a Quantus^{TM} fluorometer using the QuantiFluor® ONEdsDNAsystem (E4871) following the manufacturer's instructions. The quantified DNA was then stored at −20 for further molecular assays.

The oligonucleotides viz., MZia1:5′-TGTTATIGGTATG GTICGICGTGCTAT-3′ and CAP8: 5′-TGAAGTTACCTT CACCACGACCGG-3′ (Tétart *et al.* 2001) is used @ 1 μl each with a concentration of 0.5 pMol in the final reaction in 12·5 μl 2× Emerald GT amp master mix (TAKARA, Japan), along with 1 μl template DNA (equivalent to approx. 2–3 ng of bacteriophage genomic DNA) and PCR grade water to make up the total volume of reaction up to 25 μl. The PCR is set up with the following thermal conditions. The amplified products are run on 1·5% TAE agarose EtBr gel at 70V for 40 min, and the gel is visualized and documented using Vü gel documentation system (Pop Bio imaging, UK) (see Fig. S2).

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Conflict of Interest

None to declare.

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Author Contributions

NJ, KG, GN, GKA and KB hypothesised and designed the research plan. KB, NJ and KG drafted the article. Statistical analysis, interpretation of the data and final editing of the manuscript were made by KB, KG, NJ, GN, RVSP, AK, AKM, VBY and GKA. All the authors have reviewed the manuscript.

Data Availability Statement

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Supplementary Material